ECEN 607: Advanced Analog Circuit Tech

Lab 2: Two-Stage Amplifier Design with 3σ Driven Statistical Corner Extraction

Pre-lab or Lab Report

Name: FirstName LastName
UIN: ******Last Three Digits

Objectives:

1. Design and simulate a two-stage Miller-compensated amplifier.
2. 3σ driven statistical corner extraction for certain parameters.
3. Observe the effect of voltage and temperature variations.

Prelab:

1. A PMOS input, two-stage Miller compensated op-amp is shown below. Obtain the expressions for
   - DC Gain (A_{v0}), Dominant pole (p1), Second pole (p2), Zero (z), Gain bandwidth product (GBW), Phase Margin (PM)

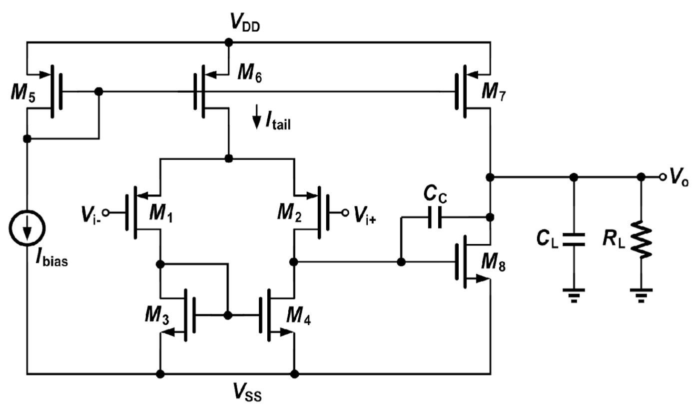


Figure 1: PMOS Two-Stage Miller Opamp

2. Design the operational amplifier of Figure 1 to obtain the following specifications:

Table 1: Design specifications

V_{DD}	0.9V
V_{SS}	-0.9V
A_{v0}	> 40 dB
GBW	> 2 MHz
PM	> 45°
Output Swing	> ±0.5 V
C_L	30 pF
R_L	10kΩ

Table 2: Reference parameters of 1.8 V NMOS transistors

Sl. No	W/L (µm/ µm)	V_{th} (mV)	µn*Cox (mF/Vs)	θ	λ (mV^{-1})	Ai (V)	Ai (dB)	f_t (GHz)
1	0.5/0.18	0.46	0.167	0.92	263	21.17	26.51	51.23
2	5/0.18	0.458	0.215	1.28	329.406	21.65	26.71	49.12
3	5/0.27	0.447	0.214	0.994	131.158	45.212	33.11	26.82

Table 3: Reference parameters of 1.8 V PMOS transistors

Sl. No	W/L (μm/ μm)	V_{tp} (mV)	μn*Cox (μF/Vs)	θ	λ (mV^{-1})	A_i (V)	A_i (dB)	f_t (GHz)
1	0.5/0.18	0.43	47.2	0.744	289.27	17.39	24.81	17.41
2	5/0.18	0.413	56.22	0.77	292.15	17.63	24.93	19.64
3	5/0.27	0.42	47.37	0.653	158.16	28.085	28.97	8.851

Lab:

3σ Driven Statistical Corner Extraction:

Corner models have been a mainstay of integrated circuit design for decades. However, there are significant inaccuracies that arise when digital CMOS corner models are used for analog circuits, or any types of circuits or measures of circuit performance they were not targeted for. The standard FF, SS SF, FS corners represent statistical bounds of global process variation for the device-level performances of speed and power. The approach is appropriate when the device-level performances of speed and power directly relate to all circuit-level performances, such as digital standard cell performances of speed and power. When these conditions are not met (i.e. when device speed/power don't directly relate to circuit output measures of interest such as DC gain, GBW etc.), then a statistical approach is more appropriate.

This part of the lab is concerned with designing circuits when global or local process variations need to be handled statistically and the target yield is 3σ. You need to extract the worst & best case 3σ corners for certain parameters.

Lab Report: Provide necessary plots, explanations etc. in the report for the following questions.

1. Simulate the design from the prelab and adjust the transistor sizes accordingly until all specifications are met.
2. Report the input referred RMS voltage noise integrated from (10 Hz – 2 MHz)
3. In addition, provide comparison tables of hand-calculated vs. final transistor sizes, and required specs (Table 1-1) vs. simulated specs. Comment on your results.
4. Run a large Monte Carlo (MC) simulation on the op-amp design. Be sure to run this simulation with process variation and mismatch. Generate histograms of the following parameters:
   a) DC Gain: A_{v0}
   b) Gain-bandwidth product: GBW
   c) Phase margin : PM

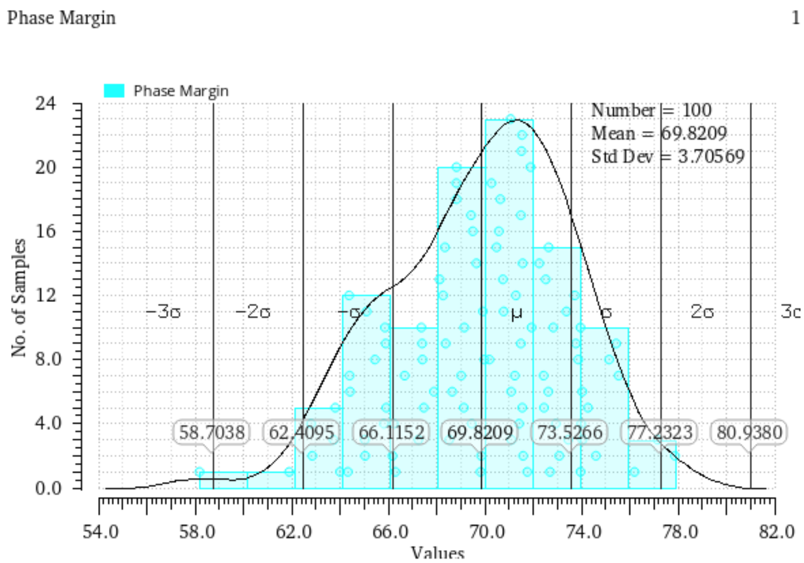


Fig 2. A sample histogram generated from MC simulation.

5. Extract the worst & best case 3σ statistical corners for A_{v0}, GBW, PM(Refer [3]) and use the option shown in Fig 3.

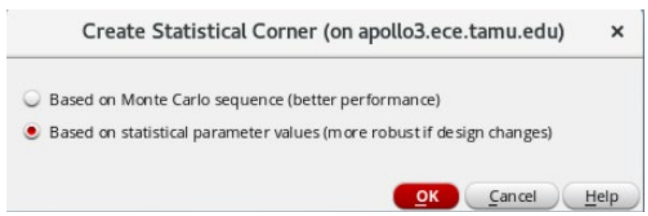
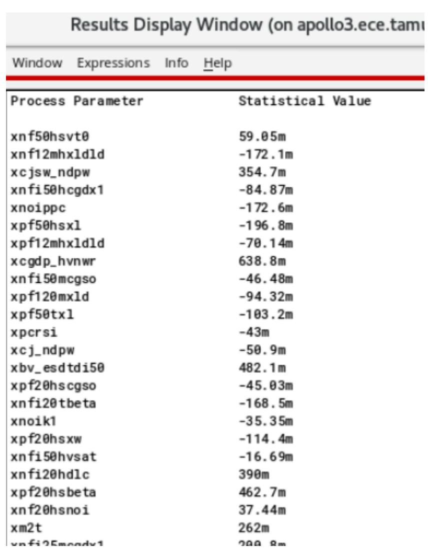


Fig 3. Creating a statistical corner



Process Parameter	Statistical Value
xnf50hsvt0	59.05m
xnf12mhxldld	-172.1m
xcjsw_ndpw	354.7m
xnfi50hcgdx1	-84.87m
xnoippc	-172.6m
xpf50hsxl	-196.8m
xpf12mhxldld	-70.14m
xcgdp_hvnwr	638.8m
xnfi50mcgso	-46.48m
xpf120mxld	-94.32m
xpf50txl	-103.2m
xpcrsi	-43m
xcj_ndpw	-50.9m
xbv_esdtdi50	482.1m
xpf20hscgso	-45.03m
xnfi20tbeta	-168.5m
xnoik1	-35.35m
xpf20hsxw	-114.4m
xnfi50hvsat	-16.69m
xnfi20hdlc	390m
xpf20hsbeta	462.7m
xnf20hsnoi	37.44m
xm2t	262m

Fig 4. Extracted corner parameter list (a sample)

6. Create a local copy of the original 1.8 V transistor model files
   a. /disk/amsc/IBM_PDK/cmhv7sf/rel/Spectre/models/nfet.scs
   b. /disk/amsc/IBM_PDK/cmhv7sf/rel/Spectre/models/pfet.scs

7. Using the 6 (3 parameters*(best + worst)) extracted corners, update the transistor models (in the copy created in part 6) manually to create two extreme corners.
   a. For this lab, please restrict within the major process parameters mentioned below. The process parameters list of 1.8 V nfet transistor is shown below. You have to go up in the hierarchy to see how these variables are assigned using the statistical parameters of Fig 4.

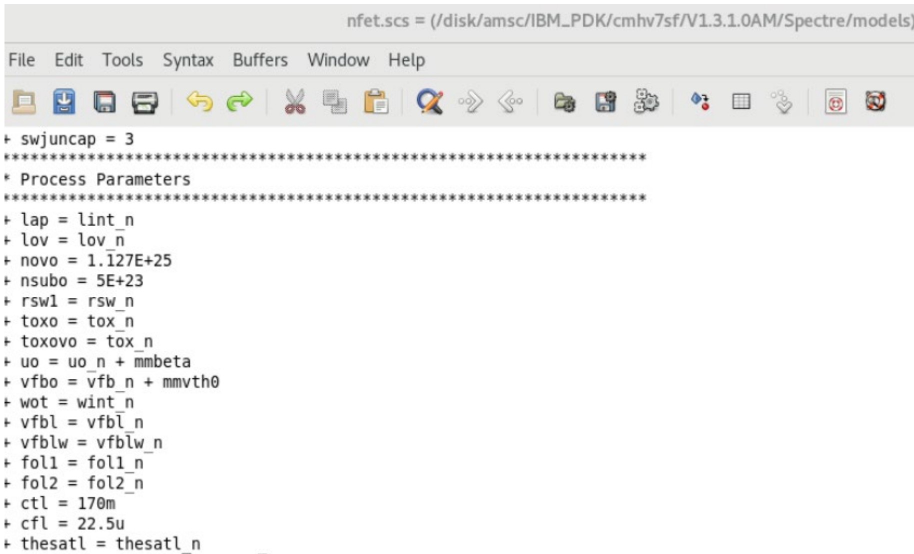


Fig 5. Process parameter list for 1.8 NFET

   b. Not all parameters of interest would be worst in the worst corner. The idea here is to observe variation of the process parameters from the different 3σ corners.
   c. Make yourself familiar with the directory/file organization of the IBM 180nm model files. For simulation, the path to the model libraries are entered in Setup->Model Libraries (shown below)

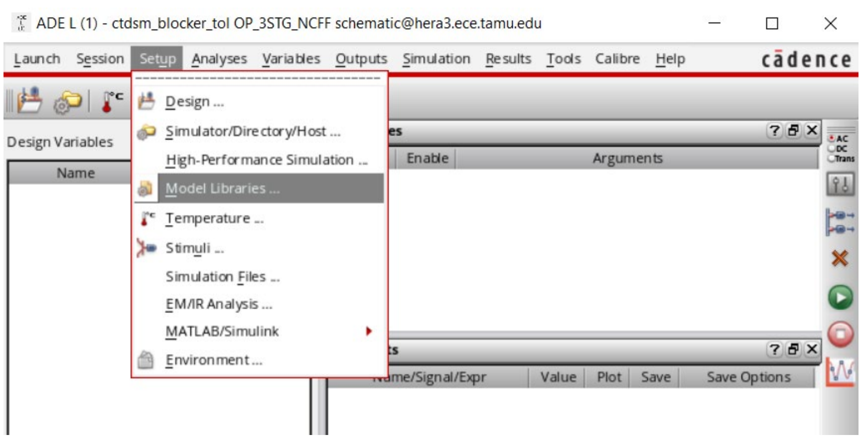


Fig 6. Adding/updating the path to the model files used in the simulation

8. Using the two extreme corners that you created, update your design so that A_{v0}, GBW, PM are within the specification. (Provide 2 plots for each parameter; one from each corner you created)
9. To sign-off the design and to verify the extracted corners, re-run a MC sim with the original models and ensure that the worst case 3σ is meeting the spec for A_{v0}, GBW, PM . (Show histograms)

10. Using the corners created, run a cross sweep simulation in ADEXL with ±10% voltage variation, -40 C < temperature < 125 C and report the worst case DC Gain, GBW and Phase Margin (you don't have to adjust the design if the specs are not met). Provide one plot for each parameter capturing the temp variation, supply variation and the corner and mark the worst case.
11. Comment on the impact of process variations and mismatch on each parameter and your overall learning from this lab.

References:

1. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=6658428
2. Variation-Aware Design of Custom Integrated Circuits: A Hands-on Field Guide, Springer 2012
3. https://community.cadence.com/cadence_blogs_8/b/cic/posts/fast-yield-analysis-and-statistical-corners